Fabric Manufacturing-II

1.**Course Code:** FE 351-0723
2. **Course Title**: Fabric Manufacturing-II.
3. **Credit**: 3.0
4. **Rationale:**

To provide students a clear concept about the basic strategies of knitting, which is one of the most popular fabric manufacturing technologies next to weaving.

5. **Course Content:**

Part: A

Knitting: Introduction and historical background of Knitting Technology, Raw materials for knitting. General terms and principles of knitting Technology; Knitting action of latch, bearded and compound needle. Basic mechanical principles of knitting Technology, elements of knitted loop structure. Basic weft knitted structures: Characteristics, Notations, Identification and uses. Basic loop or stitch type: Definition, Features and Knitting action of Tuck, Held and Float stitches.
Weft knitting machine: classification of weft knitting machinery; Plain circular latch needle machine - Description, Knitting action, Cam system, Sinker timing.
Circular Rib machine - Description, knitting action, needle timing etc.
Circular Interlock machine - Description, knitting action, interlock cam system, etc.
Flat knitting machine: Features, Classification, Main parts, Knitting action and cam system.
Calculation related to weft knitting; weft knitted fabric faults with causes and remedies.

Part: B

Warp Knitting: Introduction to warp knitting: Basic elements of warp knitting i.e. needle, sinker and guide, lapping movements of guides, pattern mechanism, warp beam, chain link, basic overlap and underlap variations, basic structures of warp knitting.
Warp knitting machinery: Classes of warp knitting machinery.
Tricot warp knitting machines: Features, knitting elements and knitting action on bearded and compound needle.
Raschel warp knitting machines: Features, knitting elements and knitting action on latch and compound needle.
Two fully threaded guide bar structures: Features, lapping diagram, chain notation and end uses
Calculation related to warp knitting. Warp knitted fabric faults.
Non-woven: Introduction to nonwoven, Methods of **web formation**. Web bonding techniques, economic advantages, **uses** of non-woven fabrics.

6. **Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)**

No	CLO Statement	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12

CLO1	Identify &Illustrate different knitting elements and Explain their role in knitting.	✓											
CLO2	Sketch &Demonstrate the knitting principles of weft and warp knitting machines for various structures			✓									
CLO3	Point out different knitting & nonwoven faults, their reasons and remedies.		✓										
CLO4	Apply latest engineering techniques in producing different knit & nonwoven fabrics..	✓											
CLO5	Construct appropriate cam, needle and link arrangement to produce particular knitted structure.					✓							

7. **Course plan specifying content, CLOs, co-curricular activities (if any), and assessment strategy mapped with CLOs.**

Lesson	**Topic**	**Teaching Learning Strategy**	**Assessment Strategy**	**Corresponding CLO**
Part- A				
1	Introduction and historical background of knit fabric manufacturing. General terms related to knitting, characteristics of knitting yarn.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO1**
2	Brief discussion about elements of knitting & their functions.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO1**
3	Knitting Action of different types of needles.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO1**
4	Basic weft knitted structures & their properties	Multimedia Presentation	Verbal interaction, sudden test, class	**CLO4**

		(MMP) and Whiteboard, Online platform, Video	test, Final examination.	
5	Study on features, classification of weft-knitting machineries. Special concentration on single jersey & rib circular knitting machines.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO2**
6	Study on features, classification of interlock circular knitting machines, purl & V-bed knitting techniques & Calculations related to weft knitting.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO2, CLO4**
7	Formation of tuck stitch & float stitch, weft knitted fabric design & mostly occurred fabric defects and their remedies.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO3, CLO4, CLO5**
Part-B				
8	Introduction to warp knitting basics & general terms.	Multimedia Presentation (MMP) and Whiteboard, Online platform, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO1**
9	Broad discussion about Pattern mechanism, chain link & lapping movement.	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	**CLO4, CLO5**
10	Clear idea about Basic stitches & overlap, underlap variations in warp knitting.	Multimedia Presentation (MMP) and Whiteboard, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO1**
11	Two fully threaded guide bar warp knitted structures with Tricot warp knitting machine.	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	**CO1**
12	Features, knitting elements & stitch formation in Tricot & Raschel knitting machine equipped with bearded & compound needle.	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	**CLO2, CLO4**
13	Calculations related with warp knitting, warp knitted fabric faults & their remedies & Introduction to nonwoven,	Multimedia Presentation (MMP) and Whiteboard, Video	Verbal interaction, sudden test, class test, Final examination.	**CLO2**

14	Methods of web formation, Web bonding techniques, economic advantages, uses of non-woven fabrics	Multimedia Presentation (MMP) and Whiteboard	Verbal interaction, sudden test, class test, Final examination.	**CLO2, CLO3**

8. Assessment Plan:

Bloom's Category	CIE* (40%)			SEE** (60%)
	Attendance/ Performance (10%)	Class Test/ Quiz (20%)	Assignment/ Presentation (10%)	
Remember		25%		10%-20%
Understand	100%	50%		30%-40%
Apply		25%	50%	40%-60%
Analyse			50%	
Evaluate				
Create				

*CIE- Continuous Internal Evaluation **SEE- Semester End Examination

9. Reference Materials

- Knitting Technology- David J Spencer (Woodhead Publishing Limited, Bington Hall, Abington Cambridge CB1 6AH, England)
- Understanding Textiles for a Merchandiser- Prof. Dr. Shah Alimuzzaman
- Circular Knitting- Lyer, Mammel, Publisher-Maisenbach Bamberg
- Handbook of Nonwoven-S.J. Ruseel

Fabric Manufacturing-II Lab

1.Course Code: FE 352-0723
2. Course Title: Fabric Manufacturing-II Lab
3. Credit: 1.0
4. Rationale:

The Knitting Laboratory was designed to fit the educational needs of our students and to conduct applied research. It features circular single knit machines, double knit machines and a jacquard machine. Warp knit fabrics can be produced on tricot machine. The lab is also equipped with Flat knitting machine.

5. Course Content:

Layout of machinery involved, loop forming techniques of different circular, flat and warp knitting machines. Practical study on feeding, take down, driving and operational technique of knitting machinery, stop motion etc. detection of fabric specification from given samples.

6. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

No	CLO Statement	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	List and illustrate different knitting elements and relate their role in production	✓											
CLO2	Demonstrate the knitting action of weft and warp knitting machine when equipped with different needle		✓										
CLO3	Differentiate among different weft circular knitting machine	✓											
CLO4	Show the yarn to fabric path diagram eccentricity ratio etc.	✓											
CLO5	Criticize conventional knitting machine over modern knitting machine		✓										
CLO6	Draw the gearing diagram and calculating RPM of the machine.		✓										

7.**Course plan specifying content, CLOs, co-curricular activities (if any) and assessment strategy mapped with CLOs.**

Lesson	Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLO
1	Study on lay out plan of knitting shed.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO1**
2	Study on basic weft knitting elements.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO2**
3	Study on basic warp knitting elements	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO2**
4	Study on loop formation technique on handsocks knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO6**
5	Study on yarn to fabric path diagram in conventional single track single jersey circular knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO4**
6	Study on calculation of cylinder RPM in conventional	A brief Lecture on the	Verbal interaction, quiz, sudden test,	**CLO6**

	single track single jersey circular knitting machine.	experiment), Demonstration, provide performance sheet, performing experiment by the students	class test, Final examination, Assignments	
7	Study on different parts & yarn to fabric path diagram of modern four track single jersey circular knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO1**
8	Study on rib circular knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO1**
9	Study on knitting action & needle timing of rib circular knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO1, CLO2**
10	Study on yarn to fabric path diagram in conventional interlock circular knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO4**
11	Study on calculation of cylinder & dial RPM in conventional interlock circular knitting machine	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO6**

		experiment by the students		
12	Study on modern interlock circular knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO1**
13	Study on yarn to fabric path diagram in tricot warp knitting machine.	A brief Lecture on the experiment), Demonstration, provide performance sheet, performing experiment by the students	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO3, CLO4**
14	Visit to a relevant manufacturing plant.	Demonstration, Group discussion, case study	Verbal interaction, quiz, sudden test, class test, Final examination, Assignments	**CLO1, CLO5**

8. Assessment Plan

<table>
<tr><th rowspan="2">Bloom's Category</th><th colspan="3">CIE* (60%)</th><th colspan="2">SEE** (40%)</th></tr>
<tr><th>Lab Attendance (10%)</th><th>Experiment and Performance (40%)</th><th>Lab Report (10%)</th><th>Experiment (20%)</th><th>Viva (20%)</th></tr>
<tr><td>Cognitive domain</td><td>10</td><td></td><td></td><td></td><td rowspan="2">20</td></tr>
<tr><td>Affective domain</td><td></td><td></td><td>10</td><td></td></tr>
<tr><td>Psychomotor domain</td><td></td><td>40</td><td></td><td>20</td><td></td></tr>
</table>

*CIE- Continuous Internal Evaluation **SEE- Semester End Examination

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